Lakshmi Narain College of Technology, Bhopal

Assignment No. 1:- Roll No. ..... Scholar No. .....

CO-1

Case Study 1:-

Q.1.1:- convert the following decimal numbers into binary:-

a. 53.625

2	53	1
2	26	0
2	13	1
2	6	0
2	3	1
	1	

$0.625 \times 2 = 1.25$ — 1

$0.25 \times 2 = 0.5$ — 0

$0.5 \times 2 = 1.0$ — 1

$(53.625)_{10} = (110101.101)_2$

b. 255.0625

2	255	1
2	127	1
2	63	1
2	31	1
2	15	1
2	7	1
2	3	1
	1	

$0.0625 \times 2 = 0.1250$ — 0

$0.125 \times 2 = 0.250$ — 0

$0.25 \times 2 = 0.5$ — 0

$0.5 \times 2 = 1.0$ — 1

$\therefore (255.0625)_{10} = (11111111.0001)_2$

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Real part is converted by successively dividing by 2 and write the remainder from bottom to top whereas the fractional part is converted by successively multiplying by 2 and write the part from top to bottom

Q.1.2

The processor receives the following binary data. convert each number into its decimal equivalent:

(a) 110110.0011

$\overleftarrow{5\ 4\ 3\ 2\ 1\ 0}$ $\overrightarrow{1\ 2\ 3\ 4}$

$110110 . 0011$

$= 1\times2^5 + 1\times2^4 + 0 + 1\times2^2 + 1\times2 + 0 + 0 + 0 + 1\times2^{-3} + 1\times2^{-4}$

$= 32 + 16 + 4 + 2 + \frac{1}{8} + \frac{1}{16}$

$= 54 + 0.1250 + 0.0625$

$= 54.1875$

$(110110.0011)_2 = (54.1875)_{10}$

(b) $\overleftarrow{5\ 4\ 3\ 2\ 1\ 0}$ $\overrightarrow{1\ 2\ 3\ 4\ 5}$

$(101110 . 11011)_2$

$= 1\times2^5 + 0 + 1\times2^3 + 1\times2^2 + 1\times2 + 0 + 1\times2^{-1} + 1\times2^{-2} + 0 + 1\times2^{-4} + 1\times2^{-5}$

$= 32 + 8 + 4 + 2 + \frac{1}{2} + \frac{1}{4} + \frac{1}{16} + \frac{1}{32}$

$= (46.84375)_{10}$

$(101110 . 11011)_2 = (46.84375)_{10}$

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Q. 1.3

For displaying numerical information on different interfaces, convert the following decimal values into octal and hexadecimal:

(a) 564.886

8	564	4 ↑
8	70	6
8	8	0
	1	

16	564	4 ↑
16	35	3
	2	

0.886 × 8 = 7.088	7	↓
0.088 × 8 = 0.704	0	
0.704 × 8 = 5.632	5	
0.632 × 8 = 5.056	5	

0.886 × 16 = 14.176	14 (E)
0.176 × 16 = 2.816	2
0.816 × 16 = 13.056	13 (D)
0.056 × 16 = 0.896	0

$(564.886)_{10} = (1064.7055\ldots)_8 = (234.E2D0\ldots)_{16}$

(b) 255.654

8	255	7 ↑
8	31	7
	3	

0.654 × 8 = 5.232	5
0.232 × 8 = 1.856	1
0.856 × 8 = 6.848	6 ↓

$(255.654)_{10} = (377.516\ldots)_8$

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16 | 255	15(F) ↑	0.654 × 16 = 10.464 — 10 (A)
15(F)		0.464 × 16 = 7.424 — 7
		0.424 × 16 = 6.784 — 6
		0.784 × 16 = 12.544 — 12 (C) ↓

$(255.654)_{10} = (FF.A7C\ldots)$

Q.1.4

The communication interface receives binary data in the following format. Convert each number into octal and hexadecimal

(a) 011010101.110011010

011 010 101 . 110 011 010

3 2 5 . 6 3 2

$(110101010.110011101)_2 = (325.632)_8$

110101010.110011101

1101 0101 . 1100 1101

13(D) 5 . 12(C) 13(D)

$(110101010.110011101)_2 = (D5.CD)_{16}$

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6. 1100.0111

001100.011100

001 100 011 100

1 4 3 4

$(1100.0111)_2 = (14.34)_8$

1100.0111

1100 0111

12 (C) 7

$(1100.0111)_2 = (C.7)_{16}$

$= (C.7)_{16}$

Q.1.5

The controller uses BCD for decimal display and Gray code for position sensing.

(a) Convert 456.78 into BCD

$(456.78)_{10} \Rightarrow$ 4 5 6 7 8

0100 0101 0110 0111 1000

$(456.78)_{10} = (0100\ 0101\ 0110 . 0111\ 1000)_{BCD}$

(b) Convert (125.15) into BCD:

$(125.15)_{10}$ 1 2 5 . 1 5

0001 0010 0101 0001 0101

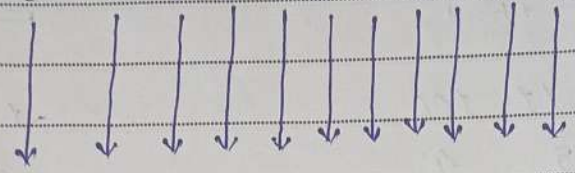
$(125.15)_{10} = (0001\ 0010\ 0101 . 0001\ 0101)_{BCD}$

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c. Convert Grey code from 11001101111

1 1 0 0 1 1 0 1 1 1 1

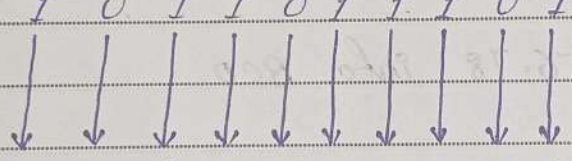


(1) 0 1 0 1 0 1 1 0 0 0

$(11001101111)_2 = (10101011000)_{grey}$

d. Convert 10110111101 into Grey code.

1 0 1 1 0 1 1 1 1 0 1



(1) 1 1 0 1 1 0 0 1 1

$(10110111101)_2 = (11101100111)_{grey}$

(e) Explain why BCD and Grey code are useful in digital system.

BCD:-

- Useful in digital clocks and calculators.
- Useful in decimal displays.
- Makes decimal digit representation easy.

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Grey Code:
- Uses in position / shaft encoders.
- only one bit changes at a time between consecutive values.
- Reduces errors during mechanical switches.

Q.1.6
Compare Binary, BCD, and Grey Code with respect to representation, conversion requirements, applications and advantages.

Basis	Binary Code	BCD code	Gray code.
Representation:	Represents number in base 2 (0,1)	Each decimal digit is represented by 4 bits	Adjacent number differ by only 1 bit
Conversion:	Decimal to binary using repeated division by 2.	Convert each decimal digit separately into 4-bit binary.	Binary to Gray: MSB remains same; XOR adjacent binary bits.
Applications:	Computers, arithmetic operations, digital circuits.	Digital clocks, calculators, decimal displays	Rotary encoders, position sensors, error reduction.
Advantages:	Simple, efficient arithmetic used in digital systems.	Easy decimal representation & display	Minimize bit changes & reduces transition errors.
Eg:	101	0101	0111

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Case Study 2:-

Q.2.1:-

Design an XOR gate using only NAND gates and explain its operation

Sol:- XOR gate:-

Output = $A \oplus B = A \cdot \bar{B} + \bar{A} \cdot B$

Truth Table:

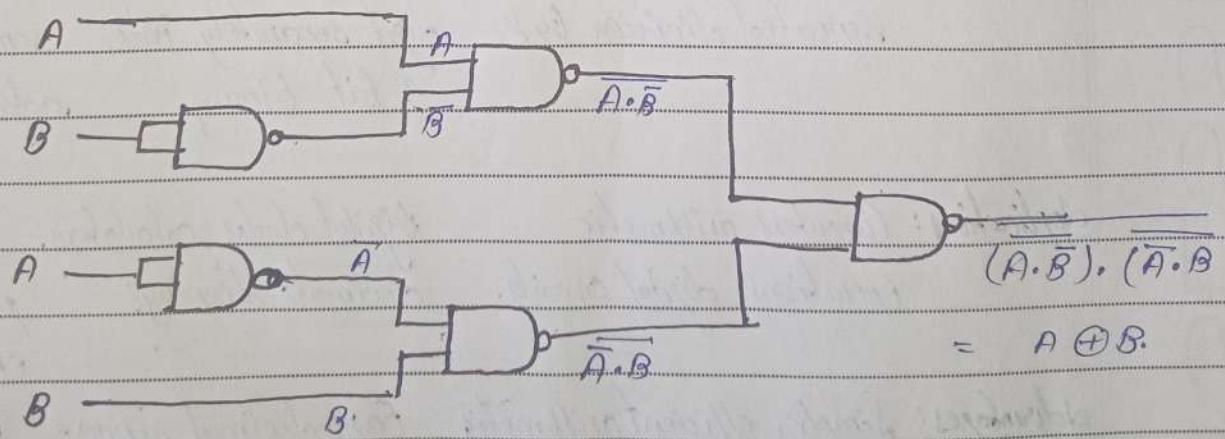
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Now,

$A \cdot \bar{B} + \bar{A} \cdot B$ as we know in NAND only (.) operator will be used, we will remove (+).

$= \overline{A \cdot \bar{B} + \bar{A} \cdot B}$

$= \overline{(A \cdot \bar{B})} \cdot \overline{(\bar{A} B)}$



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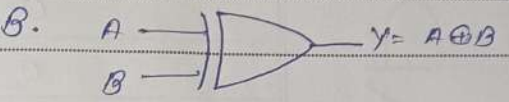
Q.2.2

Design an XOR gate using only NOR gates also design an XNOR gate using only NAND gates and only NOR gates.

- NOR ——— XOR

Output of XOR = $A \oplus B = A \oplus B$.

$= A \cdot \bar{B} + \bar{A} \cdot B$



Truth Table:

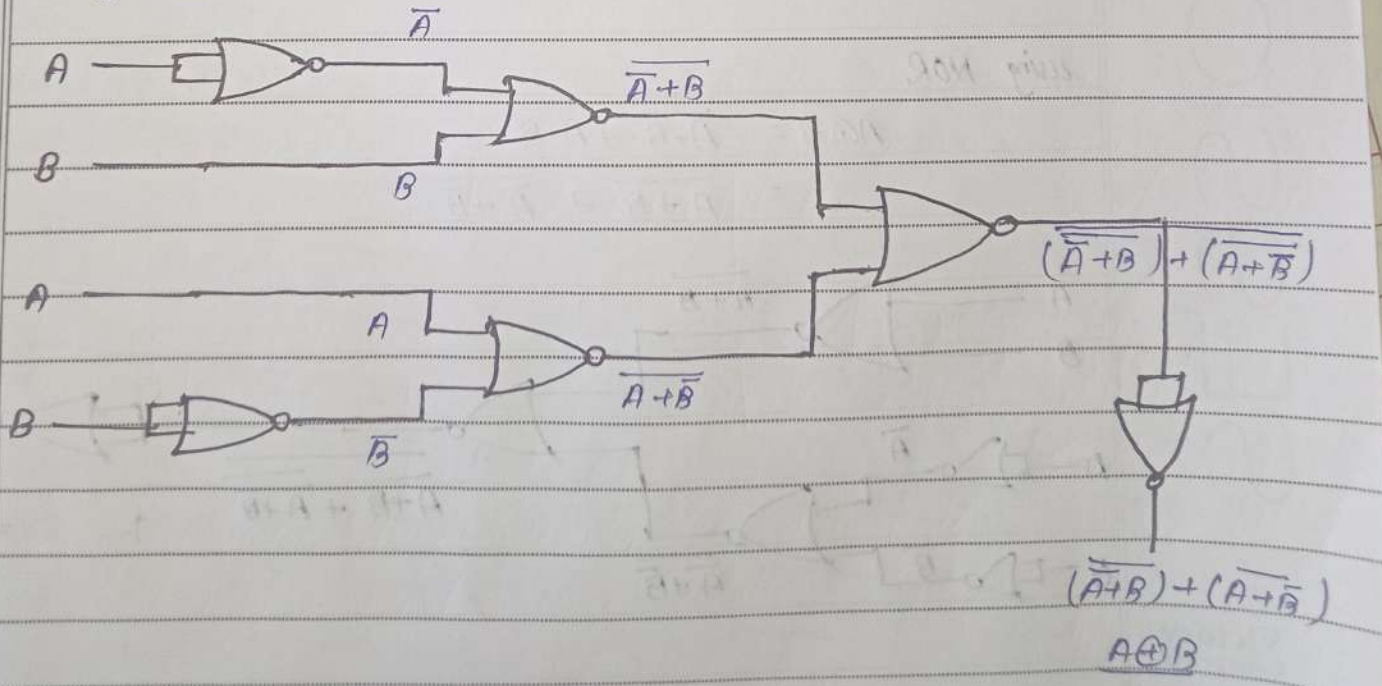
A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

$A \oplus B = A \cdot \bar{B} + \bar{A} \cdot B$

$= \overline{\overline{A \cdot \bar{B}} + \overline{\bar{A} \cdot B}}$

$= \overline{\bar{A} + B} + \overline{A \cdot \bar{B}}$

using (NOR)



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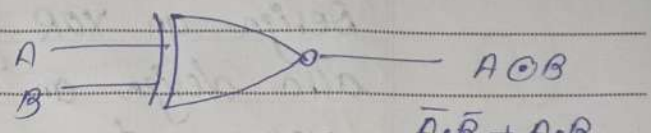
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NAND ⟶ XNOR

Output of XNOR = $A \odot B$

$= \bar{A}\cdot\bar{B} + A\cdot B$



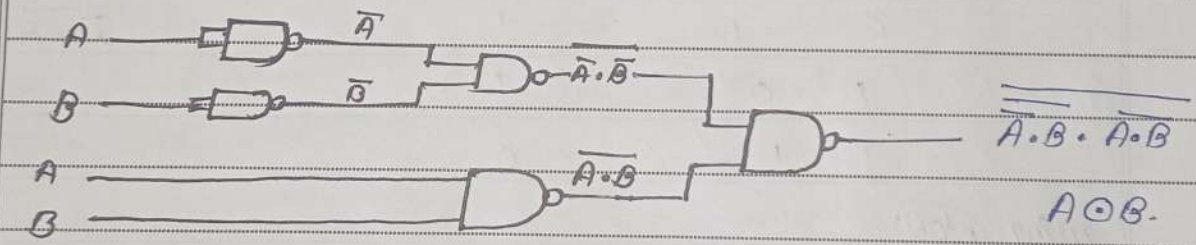
Truth Table:

A	B	A⊙B
0	0	1
0	1	0
1	0	0
1	1	1

$A \odot B = \overline{\bar{A}\cdot\bar{B} + A\cdot B}$

$= \overline{(\bar{A}\cdot\bar{B}} \cdot \overline{(A\cdot B)}$

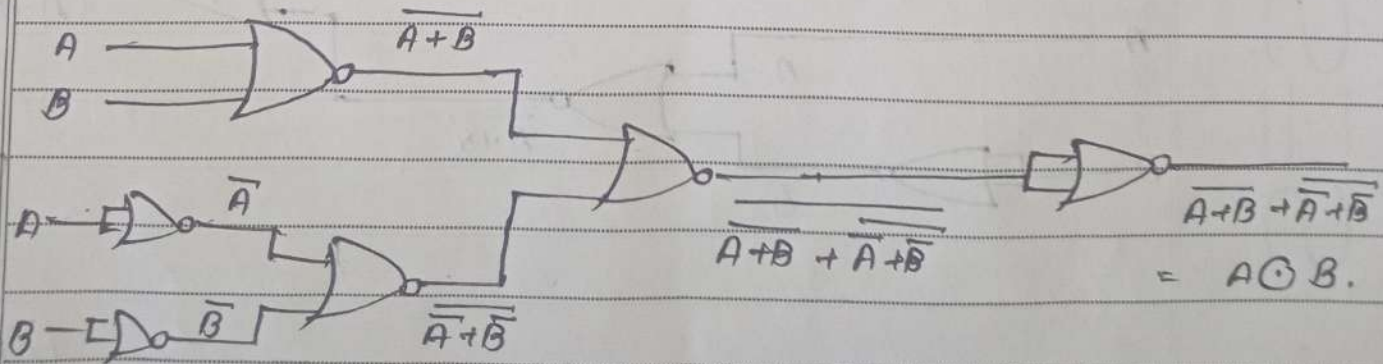
using NAND.



using NOR

$A \odot B = \overline{\bar{A}\cdot\bar{B}} + \overline{A\cdot B}$

$= \overline{A+B} + \overline{\bar{A}+\bar{B}}$



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Q. 2.3

A four variable control func^n is given by,

$f(A,B,C,D) = \Sigma(0,1,2,3,4,6,8,9,10,11)$

using a 4-variable K-map.

(a) obtain the min^m SOP expression.

(b) obtain the min^m POS expression

(c) Draw the corresponding logic circuits.

4):

$f(A,B,C,D) = \Sigma(0,1,2,3,4,5,8,9,10,11)$

(a)

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1 (0)	1 (1)	1 (3)	1 (2)
$\bar{A}B$	1 (4)	(5)	(7)	1 (6)
AB	(12)	(13)	(15)	(14)
$A\bar{B}$	1 (8)	1 (9)	1 (11)	1 (10)

$f(A,B,C,D) = \bar{B} + \bar{A}\bar{D}$

(b)

AB \ CD	$\bar{C}+\bar{D}$	$\bar{C}+D$	$C+D$	$C+\bar{D}$
$A+B$	0	1	3	2
$A+\bar{B}$	4	0 (5)	0 (7)	6
$\bar{A}+\bar{B}$	0 (12)	0 (13)	0 (15)	0 (14)
$\bar{A}+B$	8	9	11	10

$f = \Pi(5,7,12,13,14,15)$

$f(A,B,C,D) = (\bar{A}+\bar{B}) \cdot (\bar{B}+\bar{D})$

$= (\bar{A}+\bar{B}) \cdot (\bar{B}+\bar{D})$

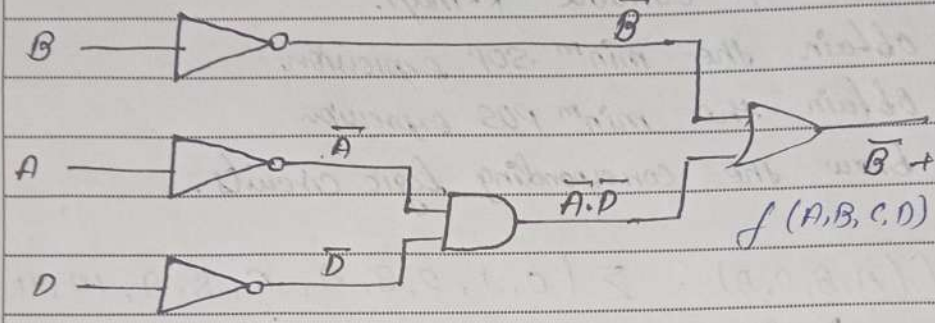
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(c)

$f(A,B,C,D) = \overline{B} + \overline{A}\,\overline{D}$

$f(A,B,C,D) = (\overline{A}+\overline{B}) \cdot (\overline{B}+\overline{D})$



$f(A,B,C,D) = \overline{B} + \overline{A}\cdot\overline{D}$

Q. 2. :

The output of a safety-control circuit is:

$Y = A\cdot B + B\cdot C + C\cdot A$

Obtain the canonical SOP form of the funcⁿ.

$Y = A\cdot B + B\cdot C + C\cdot A$

$Y = A\cdot B\cdot 1 + B\cdot C\cdot 1 + C\cdot A\cdot 1$ (AND law)

$Y = A\cdot B(C+\overline{C}) + B\cdot C(A+\overline{A}) + C\cdot A\cdot(B+\overline{B})$ (OR law)

$Y = A\cdot B\cdot C + A\cdot B\cdot\overline{C} + B\cdot C\cdot A + B\cdot C\cdot\overline{A} + C\cdot A\cdot B + C\cdot A\cdot\overline{B}$

(Distributive law)

$Y = A\cdot B\cdot C + A\cdot B\cdot\overline{C} + A\cdot B\cdot C + \overline{A}\cdot B\cdot C + A\cdot B\cdot C + A\cdot\overline{B}\cdot C$

(Commutative AND law)

$Y = A\cdot B\cdot C + \overline{A}\cdot B\cdot C + AB\overline{C} + A\cdot B\cdot\overline{C}$ (Commutative OR law)

Hence,

Canonical SOP form, $Y = A\cdot B\cdot C + \overline{A}\cdot B\cdot C + A\cdot\overline{B}\cdot C + A\cdot B\cdot\overline{C}$

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Q.2.5

Another control output is represented by:

$Y = (A+B).(B+C).(C+A)$

Obtain the canonical POS form.

Ans: $Y = (A+B).(B+C).(C+A)$

$= (A+B+1).(B+C+1).(C+A+1)$ (OR law)

$= [A+B+(C.\bar{C})].(B+C+A.\bar{A}).(C+A+B.\bar{B})$ (and law)

$= (A+B+C).(A+B+\bar{C}).(B+C+A)(B+C+\bar{A}).(C+A+B).(C+A+\bar{B})$ (distributive law)

$= (A+B+C)(A+B+\bar{C})(A+B+C)(\bar{A}+B+C)(A+B+C)(A+\bar{B}+C)$ (commutative OR law)

$Y = (A+B+C)(\bar{A}+B+C)(A+\bar{B}+C)(A+B+\bar{C})$ (commutative AND law)

→ Canonical POS form

Q.2.6:

Explain how Boolean minimization can reduce in a practical digital control system.

(a) Number of gates:

Eg: original $Y = AB + AB'$

$Y = A(B+B')$ (using boolean expression)

$Y = A.1$

$Y = A$

- before minimization: AND, NOT and OR gates are required
- after minimization: No logic gate is required;
- Result: Reduces the number of gates and simplifies the circuit.

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b. Number of inputs:

as. $Y = AB + AB'$.

$Y = A.(B+B')$.

$Y = A.1$

$Y = A$.

- Before minimization : Two inputs, A and B are required.
- After minimization : Only input A is required.
- Result : Reduces the number of input signals and simplifies circuit connections.

c. propagation delay:

$Y = AB + AB'$

$Y = A$.

- Before minimization : The signal passes through multiple logic gates, causing delay.
- After minimization : The input A is directly connected to output Y.

Result : Reduces propagation delay and Its circuit speed.

d. Hardware cost:

$Y = AB + AB'$, $Y = A$

Before minimization: More logic gates, wiring and power are required.
After minimization: No logic gates are required; only direct connecti...
Result: Hardware cost Reduces, power consumption reduces and circuit size reduces.

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